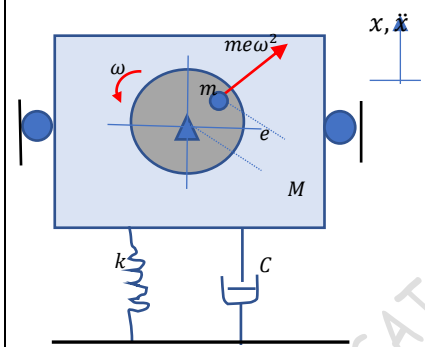


Rotating Unbalance

Consider a machine of mass M resting on a base with stiffness k and damping coefficient c . The machine has a rotating element of mass m whose centre of mass has an eccentricity e and is rotating at an angular velocity ω .

The unbalanced force $= me\omega^2$. Assuming that the machine is rigidly fastened to the base, the horizontal component of the unbalanced force $me\omega^2 \cos \theta$ will be balanced by the support. But the vertical component $me\omega^2 \sin \theta$ will cause the system to vibrate in the vertical direction.

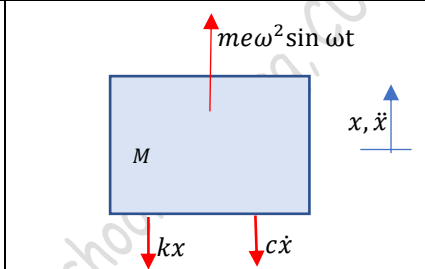


From the free body diagram, the equation of motion of the machine can be written as

$$M\ddot{x} = me\omega^2 \sin \omega t - c\dot{x} - kx$$

$$M\ddot{x} + c\dot{x} + kx = me\omega^2 \sin \omega t$$

The governing equation is a second order, non homogeneous, linear, ordinary differential equation with constant coefficients. The solution will have two parts; a homogenous solution and a particular solution



The homogeneous solution is obtained from the homogeneous part $m\ddot{x} + c\dot{x} + kx = 0$

Homogeneous solution:

$$x_h(t) = (A_1 + A_2 t)e^{-\zeta\omega_n t}, \quad \text{if } \zeta = 1$$

$$x_h(t) = e^{-\zeta\omega_n t} (A_1 e^{\omega_n \sqrt{\zeta^2 - 1} t} + A_2 e^{-\omega_n \sqrt{\zeta^2 - 1} t}), \quad \text{if } \zeta > 1$$

$$x_h(t) = X_0 e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi), \quad \text{if } \zeta < 1$$

Particular solution:

The particular solution will be of the similar form as $\frac{F_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$

$$x_p(t) = \frac{\frac{me\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$$

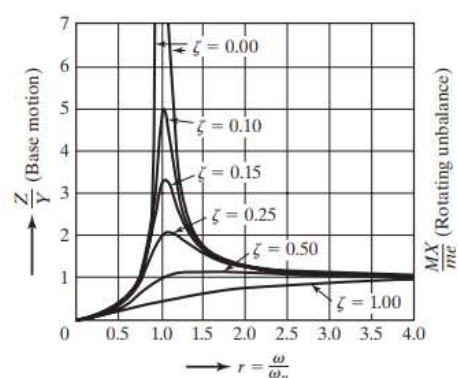
$$x_p(t) = \frac{\frac{me\omega^2}{M\omega_n^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta) \quad \ll \gg \quad \text{where} \quad \tan \theta = \frac{2r\zeta}{1-r^2}$$

At steady state, the amplitude of response of the system,

$$X = \frac{\left(\frac{me}{M}\right)r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

$$\frac{X}{\left(\frac{me}{M}\right)} = \frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

To find the maximum of $\frac{X}{\left(\frac{me}{M}\right)}$, $\frac{d}{dr} \left\{ \frac{X}{\left(\frac{me}{M}\right)} \right\} = 0$



$$\frac{r^2 \times \frac{1}{2\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \times \{2(1-r^2)(-2r) + 4\zeta^2(2r)\} - \sqrt{(1-r^2)^2 + (2r\zeta)^2} \times (2r)}{\{(1-r^2)^2 + (2r\zeta)^2\}^{3/2}} = 0$$

$$r = \frac{1}{\sqrt{1-2\zeta^2}} \quad \left(0 < \zeta < \frac{1}{\sqrt{2}}\right) \quad \ll \gg \quad \frac{X}{\left(\frac{me}{M}\right)} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

Force transmitted to the base

The force transmitted to the base, $f_T = c\dot{x} + kx$

Assuming steady state, $x(t) = \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta) \ll \gg \dot{x}(t) = \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \omega \cos(\omega t - \theta)$

$$f_T = c\omega \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \cos(\omega t - \theta) + k \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$$

$$f_T = \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \{c\omega \cos(\omega t - \theta) + k \sin(\omega t - \theta)\}$$

$$f_T = \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} A \{\sin \psi \cos(\omega t - \theta) + \cos \psi \sin(\omega t - \theta)\}$$

$$c\omega = A \sin \psi \ll \gg k = A \cos \psi$$

$$A = \sqrt{k^2 + (c\omega)^2} \ll \gg A = k \sqrt{1 + \left(\frac{c\omega}{k}\right)^2}$$

$$A = k \sqrt{1 + (2r\zeta)^2} \ll \gg \tan \psi = \frac{c\omega}{k} = \frac{2r\zeta}{1-r^2}$$

$$f_T = \frac{\frac{m\omega^2}{k}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} k \sqrt{1 + (2r\zeta)^2} \sin(\omega t - \theta + \psi) = \frac{m\omega^2 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta + \psi) = \frac{m\omega^2 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \alpha)$$

The amplitude of force transmitted to the base at steady state,

$$F_T = \frac{m\omega^2 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The ratio of amplitude of dynamic force transmitted to the base to the amplitude of excitation force is known as **force transmissibility** β_f

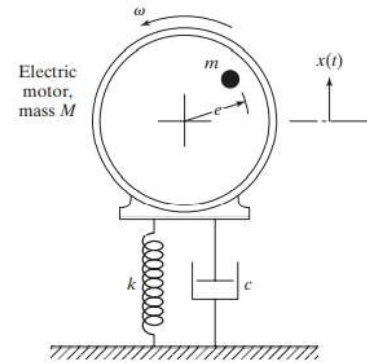
$$\beta_f = \frac{F_T}{m\omega^2} = \frac{\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

The phase angle $\alpha = \theta - \psi \ll \gg \tan \alpha = \frac{\tan \theta - \tan \psi}{1 + \tan \theta \tan \psi} = \frac{\frac{2r\zeta}{1-r^2} - 2r\zeta}{1 + \frac{2r\zeta}{1-r^2} 2r\zeta} = \frac{2\zeta r^3}{1-r^2+4\zeta^2 r^2}$

#Example 1

An electric motor of mass M , mounted on an elastic foundation, is found to vibrate with a deflection of 0.15 m at resonance. It is known that the unbalanced mass of the rotor is 8% of the mass of the motor due to manufacturing tolerances used, and the damping ratio of the foundation is $\zeta = 0.025$. Determine the following:

- 1) Eccentricity of the the unbalanced mass,
- 2) The peak deflection of the motor when the frequency ratio varies from resonance, and
- 3) The additional mass to be added uniformly to the motor if the deflection of the motor at resonance is to be reduced to 0.1 m . Assume that the eccentric mass remains unaltered when the additional mass is added to the motor.



$$X = 0.15\text{ m}, \quad r = 1, \quad \zeta = 0.025, \quad m = 0.08M$$

$$\frac{X}{\left(\frac{me}{M}\right)} = \frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = \frac{1}{2\zeta} = 20$$

$$\frac{me}{M} = \frac{0.15}{20} = 0.0075 \gg e = 0.09375\text{ m}$$

Peak deflection outside resonance,

$$\left\{ \frac{X}{\left(\frac{me}{M}\right)} \right\}_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} = 20.0063 \gg X_{\max} = 20.0063 \times 0.08 \times 0.09375 = 0.150047\text{ m}$$

The additional mass to be added to the motor to reduce the deflection to 0.1 m at resonance

$$\frac{0.1}{\left(\frac{0.08M \times 0.09375}{M+a}\right)} = \frac{1}{2\zeta} = 20 \gg 0.1(M+a) = 0.15M$$

$$a = 1.5M - M = 0.5M$$

Exercise 1

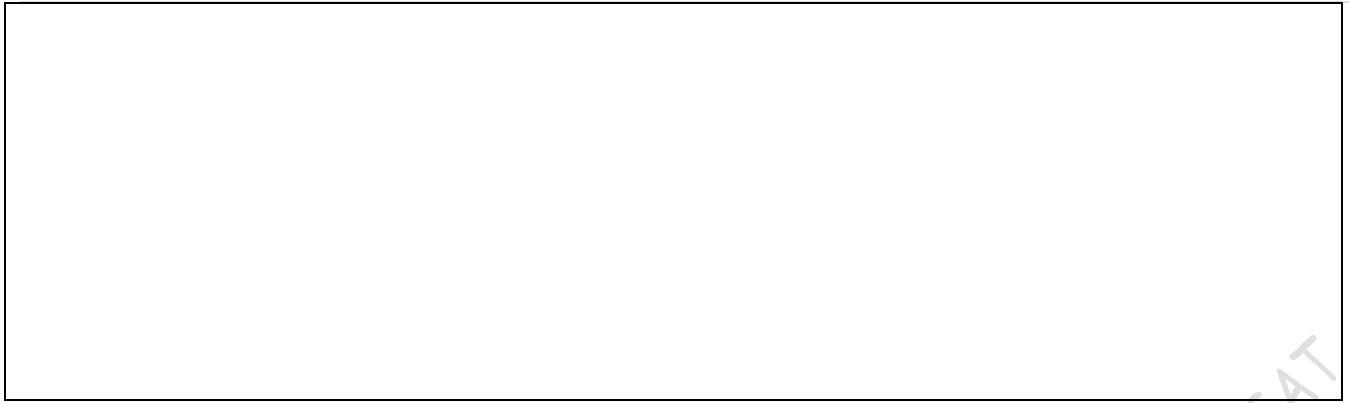
A turbine rotor of mass 200 kg has an unbalanced mass of 15 kg . It is supported on a foundation which has an equivalent stiffness of 5000 N/m and a damping ratio of $\zeta = 0.05$. If the rotor is found to vibrate with a deflection of 0.1 m at resonance, determine (a) the eccentricity of the unbalanced mass, (b) additional mass to be added uniformly to the rotor if the deflection of the rotor at resonance is to be reduced to 0.05 m . (c) peak deflection of the turbine outside resonance

Example 2

A rotating machine of mass $M = 100\text{ kg}$ is supported on four elastic mounts, each having a stiffness of $k = 50,000\text{ N/m}$ and a damping constant of $c = 500\text{ N.s/m}$. The machine rotates at a speed of 6000 rpm and has an eccentric mass $m = 0.005\text{ kg}$ at a distance of $e = 0.1\text{ m}$ from the axis of rotation. Determine the amplitude of vibration of the machine.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition



Vibration & Noise Control..... Dr. Biju N., Professor, School of Engg, CUSAT